ONLINE COURSE ENROLLMENT

ASSIGNMENT PROMPTS

PROBLEM STATEMENT: ONLINE COURSE ENROLLMENT

Design an Online Course Enrollment System to manage student details, course details, instructor details, enrollment, and enrollment status. The system should allow students to view course details, select a course, enroll in the course, and check the current enrollment status.

FEATURE SET:

- Student details
- Course details
- Instructor
- Enrollment
- Enrollment status

1. PROMPT FOR ALGORITHM

Create a simple and easy-to-understand algorithm for the following college first-year assignment.

Problem Statement: Online Course Enrollment

Description: Design an Online Course Enrollment System to manage student details, course details, instructor details, enrollment, and enrollment status. The system should allow students to view course details, select a course, enroll in the course, and check the current enrollment status.

Feature Set:

• Student details

• Course details

• Instructor

• Enrollment

• Enrollment status

Requirements:

1. Write the algorithm in numbered step-by-step format.

2. Start with entering student details.

3. Display and select the required course.

4. Display the instructor details for the selected course.

5. Perform the student enrollment process.

6. Update the enrollment status.

7. Check and display the current enrollment status.

8. End the process after displaying the enrollment status.

9. Use simple language suitable for first-year students.

10. Do not add features outside the given Feature Set.

Generate the final algorithm in a clean, well-structured and easy-to-read format.

2. PROMPT FOR FLOWCHART

Create a professional, attractive and presentation-ready FLOWCHART IMAGE for a college first-year assignment.

Problem Statement: Online Course Enrollment

Feature Set: Student details • Course details • Instructor • Enrollment • Enrollment status

Create a complete logical flowchart showing the process from Start to End.

START

↓

Enter Student Details

↓

Display Course Details

↓

Select Course

↓

Display Instructor Details

↓

Enroll Student in Course

↓

Update Enrollment Status

↓

Check Enrollment Status

↓

Display Enrollment Status

↓

END

Use standard flowchart notation:

• Oval Start / End

• Parallelogram Input / Output

• Rectangle Process

• Diamond Decision, only where logically required

• Arrows Direction of flow

Visual Requirements:

• Use proper standard flowchart shapes.

• Use clean and straight connecting arrows.

• Maintain proper spacing and alignment.

• Make all text large, sharp and readable.

• Use a modern academic and professional design.

• Use subtle attractive colors while maintaining clear flowchart notation.

• Avoid overlapping shapes and lines.

• Keep the diagram simple and suitable for first-year students.

• Do not add unnecessary features.

• Add the title: "Flowchart – Online Course Enrollment"

• Generate ONLY the final high-resolution flowchart image.

3. PROMPT FOR ER DIAGRAM

Create a professional, attractive and presentation-ready ER DIAGRAM as a high-resolution image for a college first-year assignment.

Problem Statement: Online Course Enrollment

Feature Set: Student details • Course details • Instructor • Enrollment • Enrollment status

IMPORTANT: Use CLASSICAL / CHEN ER DIAGRAM NOTATION.

Use these shapes strictly:

• RECTANGLE Entity

• OVAL / ELLIPSE Attribute

• DIAMOND Relationship

• UNDERLINED ATTRIBUTE Primary Key (PK)

• Clearly indicate Foreign Keys (FK)

• Straight connector lines between entities, attributes and relationships

• Show proper cardinality near relationship lines

Entities and Attributes:

STUDENT

student_id (PK) • student_name • mobile • email • address

COURSE

course_id (PK) • course_name • duration • course_fee

INSTRUCTOR

instructor_id (PK) • instructor_name • email • specialization

ENROLLMENT

enrollment_id (PK) • student_id (FK) • course_id (FK) • enrollment_date • enrollment_status

Relationships:

• STUDENT — HAS — ENROLLMENT

• ENROLLMENT — SELECTS — COURSE

• COURSE — TAUGHT_BY — INSTRUCTOR

Visual Requirements:

• Use rectangular boxes for entities.

• Use oval shapes for attributes.

• Use diamond shapes for relationships.

• Connect every attribute to its corresponding entity.

• Connect entities through relationship diamonds.

• Clearly show Primary Keys and Foreign Keys.

• Show appropriate cardinality such as 1:1 or 1:M.

• Keep the diagram balanced and well organized.

• Avoid crossing connector lines wherever possible.

• Use a clean white background.

• Use sharp, readable text.

• Use subtle professional colors.

• Do not use table-style database diagrams.

• Do not use flowchart symbols.

• Do not add unnecessary entities or attributes.

• Do not use icons or 3D effects.

• No watermark.

• Add the title: "ER Diagram – Online Course Enrollment"

• Generate ONLY the final high-resolution ER diagram image.